Lecture 10

- EKF
- Unscented Kalman Filter (UKF)

Recall the EKF:

General model:

$$X_{t+1} = f(X_t, u_t) + W_t, \quad W_t \sim GWN(0, Q_t)$$
$$Y_t = g(X_t) + V_t, \quad V_t \sim GWN(0, R_t)$$

EKF Algo

Init: $X_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$

predict: ← nonlinear dynamics

$$\mu_{t|t-1} = f(\mu_{t-1|t-1}, u_{t-1})$$
$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1} \quad \text{(same as KF)}$$

update: ← nonlinear measurement model

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - g(\mu_{t|t-1}))$$
$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} \quad \text{(same as KF)}$$

where $K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$

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where A_{t-1}, C_t are Jacobians

$$A_{t-1} = \left.\frac{\partial f(x_{t-1}, u_{t-1})}{\partial x_{t-1}}\right|_{\substack{x_{t-1}=\mu_{t-1|t-1} \\ u_{t-1}=u_{t-1}}}$$

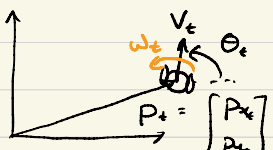
(vector index)

$$= \left.\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}\right| \in \mathbb{R}^{n\times n} \quad \left|\begin{matrix} x_{t-1}=\mu_{t-1|t-1} \\ u_{t-1}=u_{t-1} \end{matrix}\right.$$

$$C_t = \left.\frac{\partial g(x_t)}{\partial x_t}\right|_{x_t=\mu_{t|t-1}}$$

$$= \left.\begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial g_m}{\partial x_1} & \cdots & \frac{\partial g_m}{\partial x_n} \end{bmatrix}\right| \in \mathbb{R}^{m\times n} \quad \left|\, x_t=\mu_{t|t-1}\right.$$

E.g. Differential Drive Robot



state:

$$x_t = \begin{bmatrix} p_{x_t} \\ p_{y_t} \\ \theta_t \end{bmatrix}$$

CT dynamics:

$$\dot{p}_{x_t} = v_t \cos\theta_t$$
$$\dot{p}_{y_t} = v_t \sin\theta_t$$
$$\dot{\theta}_t = \omega_t$$

input:

$$u_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

DT 1st Order Euler Approx:

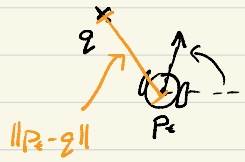
$$f(x_{t-1}, u_{t-1}) \begin{cases} p_{x_t} = p_{x_{t-1}} + \delta t\, v_{t-1}\cos\theta_{t-1} + w_{t-1}^x \\ p_{y_t} = p_{y_{t-1}} + \delta t\, v_{t-1}\sin\theta_{t-1} + w_{t-1}^y \\ \theta_t = \theta_{t-1} + \delta t\, \omega_{t-1} + w_{t-1}^\theta \end{cases}$$

where $w_{t-1} \sim \begin{bmatrix} w_{t-1}^x \\ w_{t-1}^y \\ w_{t-1}^\theta \end{bmatrix} \sim N(0, Q_{t-1})$

compute dynamics Jacobian A_{t-1}:

$$A_{t-1} = \left.\begin{bmatrix} 1 & 0 & -\delta t\, v_{t-1}\sin\theta_{t-1} \\ 0 & 1 & \delta t\, v_{t-1}\cos\theta_{t-1} \\ 0 & 0 & 1 \end{bmatrix}\right| \begin{matrix} \theta_{t-1}=\mu_{t-1|t-1}^\theta \\ v_{t-1}=v_{t-1} \end{matrix}$$

suppose we have a range measurement to a landmark



sensor model

$$Y_t = \|p_t - q\| + V_t$$

$$V_t \sim N(0, R_t)$$

compute Jacobian C_t:

$$g(x_t) = \left((p_{x_t} - q_x)^2 + (p_{y_t} - q_y)^2\right)^{1/2}$$

$$\frac{\partial g(x_t)}{\partial p_{x_t}} = \frac{1}{2}\underbrace{\left((p_{x_t} - q_x)^2 + (p_{y_t} - q_y)^2\right)^{-1/2}}_{\frac{1}{\|p_t - q\|}} \cdot 2 p_{x_t}$$

$$\frac{\partial g(x_t)}{\partial p_{y_t}} = \frac{1}{\|p_t - q\|} p_{y_t}, \quad \frac{\partial g(x_t)}{\partial \theta_t} = 0$$

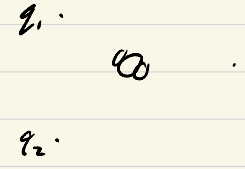
$$C_t = \left[\frac{p_{x_t}}{\|p_t - q\|} \quad \frac{p_{y_t}}{\|p_t - q\|} \quad 0\right] \Big|_{\substack{p_{x_t} = \mu^x_{t|t-1} \\ p_{y_t} = \mu^y_{t|t-1}}}$$

where $\mu_{t|t-1} = \begin{bmatrix} \mu^x_{t|t-1} \\ \mu^y_{t|t-1} \\ \mu^\theta_{t|t-1} \end{bmatrix}$

plug into EKF and ready to use!

Note: The EKF lets use track the trajectory of a diff drive robot with a single range measurement to a fixed landmark!

If we have more landmarks q^i, $i = 1, \dots, N$



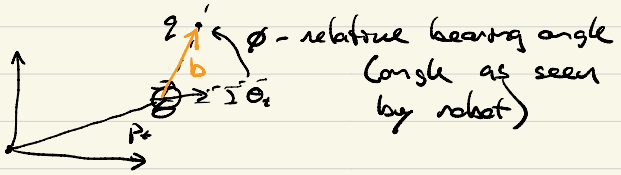
$$Y_t = \begin{bmatrix} Y^1_t \\ \vdots \\ Y^N_t \end{bmatrix}$$

Jacobian is

$$C_t = \begin{bmatrix} C^1_t \\ \vdots \\ C^N_t \end{bmatrix}$$

put in EKF update and we get sensor fusion!

consider bearing sensor:



- can consider ϕ as an angle
- or as a unit vector

unit vector:

in world coordinates:

$$b = \frac{q - p_t}{\|q - p_t\|}$$

($w \leftarrow$ world frame; $b \nearrow$ bearing)

in robot coordinate frame:

$$Y_t = \begin{bmatrix} \cos\theta_t & \sin\theta_t \\ -\sin\theta_t & \cos\theta_t \end{bmatrix} \frac{(q - p_t)}{\|q - p_t\|} + V_t$$

$$V_t \sim N(0, R_t)$$

this is $g(x_t)$

compute Jacobian C_t and apply EKF.

Towards a Derivation of EKF

consider:

$$E[X_t | Y_{1:t-1}] = E[f(X_{t-1}, u_{t-1}) + W_{t-1} | Y_{1:t-1}]$$

recall Taylor approx of $f(\cdot)$:

$$f(x_{t-1}, u_{t-1}) = f(\mu_{t-1|t-1}, u_{t-1}) + A_{t-1}(x_{t-1} - \mu_{t-1|t-1}) + \underbrace{H.O.T.s}_{\approx 0}$$

$$E[f(\cdot) | Y_{1:t-1}] \approx E[f(\mu_{t-1|t-1}, u_{t-1}) + A_{t-1}(X_{t-1} - \mu_{t-1|t-1}) + W_{t-1} | Y_{1:t-1}]$$

$$= f(\mu_{t-1|t-1}, u_{t-1}) + A_t\underbrace{\left(E[X_{t-1} | Y_{1:t-1}] - \mu_{t-1|t-1}\right)}_{= \mu_{t-1|t-1} \to 0} + \underbrace{E[W_{t-1} | Y_{1:t-1}]}_{0}$$

we find:

$$\boxed{\mu_{t|t-1} \approx f(\mu_{t-1|t-1}, u_{t-1})}$$

EKF predict

same for cov.

$$\Sigma_{t+1|t-1} = E\left[(X_t - \mu_{t|t-1})(X_t - \mu_{t|t-1})^T | Y_{1:t-1}\right]$$

plug in Taylor approx and Jacobians A_{t-1} do not disappear:

we get

$$\boxed{\Sigma_{t+1|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}} \quad (EKF)$$

same for mean and cov. predict steps.

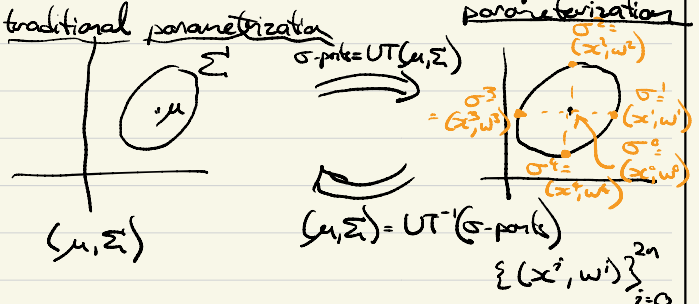
Unscented Kalman Filter (UKF)

Notice: EKF requires computation of (A_{t-1}, C_t). what if $f(\cdot), g(\cdot)$ are complex simulation, very intricate model, or have discrete logic.

UKF: Like EKF, but represents Gaussian with a set of points called σ-points

Unscented Transform

(Sigma Transform, σ-transform)



Specifically:

Forward Transform:

three parameters

$$UT(\mu,\Sigma) = \begin{cases} x^0 = \mu, & w^0 = \frac{\lambda}{\lambda+n} \\ x^i = \mu + \left(\sqrt{(\lambda+n)\Sigma}\right)_i, & w^i = \frac{1}{2(\lambda+n)}, \; i=1,\dots,n \\ x^{n+i} = \mu - \left(\sqrt{(\lambda+n)\Sigma}\right)_i, & w^{n+i} = \frac{1}{2(\lambda+n)}, \; i=1,\dots,n \end{cases}$$

where $\left(\sqrt{(\lambda+n)\Sigma}\right)_i$ is the ith column of matrix square root $\sqrt{(\lambda+n)\Sigma}$

Inverse Transform:

$$UT^{-1}(\{x^i, w^i\}_{i=0}^{2n}) = \begin{cases} \mu = \sum_{i=0}^{2n} w^i x^i \\ \Sigma = \sum_{i=0}^{2n} w^i (x^i - \mu)(x^i - \mu)^T \end{cases}$$

Verify that UT^{-1} retrieves the same (μ,Σ) used to compute $\{\sigma^i\}_{i=0}^{2n}$. That is, verify:

$$(\mu,\Sigma) = UT^{-1}(UT(\mu,\Sigma)).$$

We have

$$\mu = \sum_{i=0}^{2n} x^i w^i$$

$$= \mu \frac{\lambda}{(\lambda+n)} + \sum_{i=1}^{n} \left(\mu + \left(\sqrt{(\lambda+n)\Sigma}\right)_i\right)\frac{1}{2(\lambda+n)} + \sum_{i=1}^{n}\left(\mu - \left(\sqrt{(\lambda+n)\Sigma}\right)_i\right)\frac{1}{2(\lambda+n)}$$

$$= \mu\frac{\lambda}{\lambda+n} + \frac{n}{2(\lambda+n)}\mu + \frac{n}{2(\lambda+n)}\mu$$

$$= \mu\frac{(\lambda+n)}{(\lambda+n)} \overset{\checkmark}{=} \mu \quad \square$$

and $\Sigma = \sum_{i=0}^{2n} w^i (x^i - \mu)(x^i - \mu)^T$

$$= \frac{\lambda}{\lambda+n}\underbrace{(\mu-\mu)(\mu-\mu)^T}_{0} + \sum_{i=1}^{n}\frac{1}{2(\lambda+n)}\left(\mu + \left(\sqrt{(\lambda+n)\Sigma}\right)_i - \mu\right)\times\left(\mu + \left(\sqrt{(\lambda+n)\Sigma}\right)_i - \mu\right)^T + \sum_{i=1}^{n}\frac{1}{2(\lambda+n)}\left(\mu - \left(\sqrt{(\lambda+n)\Sigma}\right)_i - \mu\right)\times\left(\mu - \left(\sqrt{(\lambda+n)\Sigma}\right)_i - \mu\right)^T$$

$$= \sum_{i=1}^{n}\frac{1}{(\lambda+n)}\underbrace{\left(\sqrt{(\lambda+n)\Sigma}\right)_i\left(\sqrt{(\lambda+n)\Sigma}\right)_i^T}_{=\sqrt{(\lambda+n)}\sqrt{\Sigma}}$$

$$= \sum_{i=1}^{n}\left(\sqrt{\Sigma}\right)_i\left(\sqrt{\Sigma}\right)_i^T \overset{?}{=} \Sigma.$$

to see last step, notice for matrix M: $M = [m_1 \cdots m_n]$, so

$$MM^T = [m_1 \cdots m_n] \begin{bmatrix} m_1^T \\ \vdots \\ m_n^T \end{bmatrix} = \sum_{i=1}^{n} m_i m_i^T$$

for us, $\Sigma_t = (\sqrt{\Sigma^t})(\sqrt{\Sigma^t})^T$ (by def.)

$$= \sum_{i=1}^{n} (\sqrt{\Sigma^t})_i (\sqrt{\Sigma^t})_i^T \text{ as desired} \quad \square$$

We use the $UT(\cdot)$, $UT^{-1}(\cdot)$ in the UKF Algo:

UKF Algo

init: $X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$

predict:

$$UT(\mu_{t-1|t-1}, \Sigma_{t-1|t-1}) \rightarrow \{x^i_{t-1|t-1}, w^i\}_{i=0}^{2n}$$

predict each σ-point: $\boxed{\bar{x}^i_{t|t-1} = f(x^i_{t-1|t-1}, u_{t-1})}$ (no noise yet)

(w^i do not change)

$$UT^{-1}(\{\bar{x}^i_{t|t-1}, w^i\}_{i=0}^{2n}):$$

$$\boxed{\mu_{t|t-1} = \sum_{i=0}^{2n} w^i \bar{x}^i_{t|t-1}}$$

$$\boxed{\Sigma_{t|t-1} = \sum_{i=0}^{2n} w^i (\bar{x}^i_{t|t-1} - \mu_{t|t-1})(\cdots)^T + Q_{t-1}}$$

add noise cov. here.

update:

$$UT(\mu_{t|t-1}, \Sigma_{t|t-1}) \rightarrow \{x^i_{t|t-1}, w^i\}_{i=0}^{2n}$$

predict σ-point measurement: $\boxed{y^i_{t|t-1} = g(x^i_{t|t-1})}$

σ-point expected measurement: $\boxed{\hat{y}_{t|t-1} = \sum_{i=0}^{2n} w^i y^i_{t|t-1}}$

σ-point measurement covariance: $\boxed{\Sigma_y = \sum_{i=0}^{2n} w^i (y^i_{t|t-1} - \hat{y}_{t|t-1})(\cdots)^T + R_t}$

add meas. noise cov.

σ-point cross covariance: $\boxed{\Sigma_{xy} = \sum_{i=0}^{2n} w^i (x^i_{t|t-1} - \mu_{t|t-1}) \times (y^i_{t|t-1} - \hat{y}_{t|t-1})^T}$

compute Gaussian est. eqns.:

$$\boxed{\mu_{t|t} = \mu_{t|t-1} + \Sigma_{xy}\Sigma_y^{-1}(y_t - \hat{y}_{t|t-1})}$$

$$\boxed{\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}}$$

returns: $\boxed{(\mu_{t|t}, \Sigma_{t|t})}$

Notice: No Jacobians required. Only point evaluations of $f(\cdot), g(\cdot)$ functions